

PROJECT PROPOSAL

Smart Digital Closet and Outfit Recommendation System



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(APPROVAL)

Project Title: Digital Closet / Outfit Recommendation System

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Sign: _____

Problem Statement

In today's busy lifestyle, people often struggle to choose appropriate outfits that match their occasion, season, and available wardrobe. Managing a wardrobe manually is time-consuming, and individuals frequently forget what clothes they own. Common fashion mistakes include wearing mismatched colors or inappropriate footwear such as sports shoes for formal events or slippers with wedding attire.

Additionally, there is no simple digital tool available at a beginner C++ level that allows users to store their personal wardrobe digitally and receive structured outfit recommendations based on basic fashion rules.

Proposed Solution

We propose to develop a console-based Smart Digital Closet and Outfit Recommendation System in C++. The system will:

- Allow users to register and securely log in using file-based authentication.
- Let each user add three types of clothing: Upper Wear, Lower Wear, and Shoes.
- Store all data in text files so it persists between program sessions.
- Edit and Delete the Clothes.
- Generate outfit recommendations by checking occasion, season, color compatibility, and shoe-type validity rules.
- Apply strict fashion rules: formal shoes for formal/wedding occasions, sports shoes for sports, and no inappropriate footwear combinations.
- Support both Boys and Girls clothing through a gender attribute on each item.

4. Classes and Their Responsibilities

The system uses the following five classes organized in an inheritance hierarchy:

Class Name	Type	Responsibilities
ClothingItem	Base Class (Parent)	Holds common attributes: name, color, gender, season, occasion, category. Declares virtual displayItem() for polymorphism. Contains getters and helper methods.
Upper	Derived Class (Child)	Inherits ClothingItem. Adds fabric attribute for upper wear. Overrides displayItem() to print [UPPER WEAR] label.
Lower	Derived Class (Child)	Inherits ClothingItem. Adds fabric attribute for lower wear. Overrides displayItem() to print [LOWER WEAR] label.
Shoes	Derived Class (Child)	Inherits ClothingItem. Adds shoeType attribute. Overrides displayItem() to print [SHOES] label. Shoe type drives fashion rule validation.
User System (Functions)	Functional Module	Handles signup, login, file reading/writing for user accounts. Manages the current session via the currentUser global variable.

5. OOP Concepts to Be Used

OOP Concept	How It Will Be Used in This Project
Encapsulation	All class attributes are declared private/protected. Access is provided through public getter methods (getName(), getColor(), etc.) and constructors. No attribute is directly accessed from outside the class.
Inheritance	Three derived classes (Upper, Lower, Shoes) inherit from the base class ClothingItem using public inheritance. Shared attributes and methods are written once in the parent class.
Polymorphism	The displayItem() function is declared virtual in ClothingItem and overridden in Upper, Lower, and Shoes. Each class prints its own type-specific label and attributes when called.
Abstraction	The user interacts only with high-level menus (Add Clothing, View Closet, Get Recommendations). Internal logic of file parsing, color matching, and shoe validation is hidden inside functions.
File Handling	The fstream library is used to store user accounts in users.txt and clothing data in clothes.txt, ensuring data persistence between sessions.
Constructors	Each class has a parameterized constructor to initialize all attributes at the time of object creation.

6. Feasibility

This project is fully feasible within the given semester timeline. It uses only the standard C++ libraries (iostream, iomanip, string, fstream) and does not require

any external dependencies, GUI frameworks, or databases. The logic is kept at beginner OOP level with simple arrays, switch statements, and file I/O. All four team members can work on separate modules (Authentication, Upper/Lower/Shoes classes, Recommendation Engine) and integrate them into a single program.